

Chapter 6

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12:01 PM

Topics:

- Curve fitting with a linear equation

Curve Fitting With a Linear Equation

Linear least squares regression

$(0, 0.94) (30, 1.05) (70, 1.17) (100, 1.28) \quad p = a_1T + a_0$

$$S_{00} = 0 + 30 + 70 + 100 = 200$$

$$S_y = 0.94 + 1.05 + 1.17 + 1.28 = 4.44$$

$$S_{xx} = 0^2 + 30^2 + 70^2 + 100^2 = 15,800$$

$$S_{xy} = (0 \times 0.94) + (30 \times 1.05) + (70 \times 1.17) + (100 \times 1.28) = 241.4$$

$$a_1 = \frac{(4)(241.4) - (200)(4.44)}{(4)(15800) - (200)^2} = 0.003345$$

$$a_0 = \frac{(15800)(4.44) - (241.4)(200)}{(4)(15800) - (200)^2} = 0.9428$$

$$p = (0.003345)(T) + 0.9428$$

$$p(80) = (0.003345)(80) + 0.9428 = 1.2104$$

- Curve fitting with a non-linear equation by writing the equation in linear form

Curve fitting a nonlinear equation (exponential)

$p = be^{mx} \rightarrow \ln(p) = \ln(b) + mx$

$$S_x = 1900 + 1950 + 1970 + 1980 + 1990 + 2000 + 2010 = 13800$$

$$S_{xx} = 1900^2 + 1950^2 + 1970^2 + 1980^2 + 1990^2 + 2000^2 + 2010^2 = 27214000$$

$$S_y = \ln(400) + \ln(557) + \ln(825) + \ln(981) + \ln(1135) + \ln(1266) + \ln(1370) = 47.319$$

$$S_{xy} = (1900 \times \ln(400)) + (1950 \times \ln(557)) + (1970 \times \ln(825)) + (1980 \times \ln(981)) + (1990 \times \ln(1135)) + (2000 \times \ln(1266)) + (2010 \times \ln(1370)) = 121720.3182$$

$$a_1 = \frac{(7)(121720.3182) - (13800)(47.319)}{(7)(27214000) - 13800^2} \quad a_0 = \frac{(27214000)(47.319) - (121720.3182)(13800)}{(7)(27214000) - 13800^2}$$

$$a_1 = 0.003315 \quad a_0 = 2.287185$$

$$p = 9.8471788 e^{0.003315x}$$

$$e^{2.287185} = 9.8471788$$

- Curve fitting with quadratic and higher-order polynomials

Fuck this topic

- Lagrange polynomials

Lagrange Interpolating Polynomial

$$f(x) = \sum y_i L_i(x) = \sum y_i \prod \frac{(x-x_j)}{(x_i-x_j)}$$

(Lagrange functions)

$$\begin{array}{l} x: 1 \quad 2 \quad 4 \quad 5 \quad 7 \\ y: 52 \quad 5 \quad -5 \quad -40 \quad 10 \end{array}$$

$$f(x) = \frac{(x-2)(x-4)(x-5)(x-7)}{(1-2)(1-4)(1-5)(1-7)} \cdot 52 + \frac{(x-1)(x-4)(x-5)(x-7)}{(2-1)(2-4)(2-5)(2-7)} \cdot 5$$

$$+ \frac{(x-1)(x-2)(x-5)(x-7)}{(4-1)(4-2)(4-5)(4-7)} \cdot (-5) + \frac{(x-1)(x-2)(x-4)(x-7)}{(5-1)(5-2)(5-4)(5-7)} \cdot (-40)$$

$$+ \frac{(x-1)(x-2)(x-4)(x-5)}{(7-1)(7-2)(7-4)(7-5)} \cdot 10$$

Interpolate for $x=3$

$$\frac{(1)(-1)(-2)(-4)}{(-1)(-3)(-4)(-6)} \cdot 52 + \frac{(2)(-1)(-2)(-4)}{(1)(-2)(-3)(-5)} \cdot 5 + \frac{(2)(1)(-2)(-4)}{(3)(2)(-1)(-3)} \cdot (-5)$$

$$+ \frac{(2)(1)(-1)(-4)}{(4)(3)(1)(-2)} \cdot (-40) + \frac{(2)(1)(-1)(-2)}{(6)(5)(3)(2)} \cdot 10$$

$$= \frac{52}{9} + \frac{5}{3} + \frac{-40}{9} + \frac{40}{3} + \frac{2}{9} = 6$$

- Newtons polynomials

Newton's Interpolating Polynomial

$$f(x) = y_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) + a_4(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

1	52					
2	5	$\frac{5-52}{2-1} = -47$				
4	-5	$\frac{5-5}{4-1} = -47$	$\frac{-47-(-47)}{4-2} = 0$			
5	-40	$\frac{-40-5}{5-1} = -9.75$	$\frac{-9.75-(-47)}{5-2} = 12.25$			
7	10	$\frac{10-5}{7-1} = 1.67$	$\frac{1.67-(-9.75)}{7-2} = -2.14$	$\frac{-2.14-12.25}{7-4} = -4.78$		

$$y = f(x) = 52 + (-47)(x-1) + 0(x-1)(x-2) + 12.25(x-1)(x-2)(x-4) + (-4.78)(x-1)(x-2)(x-4)(x-5)$$

Newton's Interpolating Polynomial

Interpolate for $x=3$

$$y = f(3) = 52 + (-47)(3-1) + 0(3-1)(3-2) + 12.25(3-1)(3-2)(3-4) + (-4.78)(3-1)(3-2)(3-4)(3-5)$$

$$u = f(3) = 52 + (-94) + 28 + 12 + 8 = 6$$

- Piecewise (spline) interpolation

○ Linear Spline

Linear Spline

$$\begin{array}{l} x: 8 \quad 11 \quad 15 \quad 18 \\ y: 5 \quad 9 \quad 10 \quad 8 \end{array}$$

For points: 3 splines

$$f(x) = \frac{(x-x_2)}{(x_1-x_2)} y_1 + \frac{(x-x_1)}{(x_2-x_1)} y_2 = \frac{(x-11)}{(8-11)} 5 + \frac{(x-8)}{(11-8)} 9$$

Linear Splines

$x: 8 \quad 11 \quad 15 \quad 18$
 $y: 5 \quad 9 \quad 10 \quad 8$
 For points $\therefore 3$ splines

$$f_1(x) = \frac{(x-x_2)y_1}{(x_1-x_2)} + \frac{(x-x_1)y_2}{(x_2-x_1)} = \frac{(x-11)5}{(8-11)} + \frac{(x-8)9}{(11-8)}$$

$$= -\frac{5}{3}(x-11) + 3(x-8) \text{ for } 8 \leq x \leq 11$$

$$f_2(x) = \frac{(x-x_3)y_2}{(x_2-x_3)} + \frac{(x-x_2)y_3}{(x_3-x_2)} = \frac{(x-15)9}{(11-15)} + \frac{(x-11)10}{(15-11)}$$

$$= -\frac{9}{4}(x-15) + \frac{10}{4}(x-11) \text{ for } 11 \leq x \leq 15$$

$$f_3(x) = \frac{(x-x_4)y_3}{(x_3-x_4)} + \frac{(x-x_3)y_4}{(x_4-x_3)} = \frac{(x-18)10}{(15-18)} + \frac{(x-15)8}{(18-15)}$$

$$= -\frac{10}{3}(x-18) + \frac{8}{3}(x-15) \text{ for } 15 \leq x \leq 18$$

The interpolated value for $x=12.7 = f_2(12.7) = -\frac{9}{4}(12.7-15) + \frac{10}{4}(12.7-11) = 9.425$

○ Quadratic Spline

Spline Interpolation

$x: 8 \quad 11 \quad 15 \quad 18 \quad 22$
 $y: 5 \quad 9 \quad 10 \quad 8 \quad 7$
 There are $n=5$ points, thus 4 splines
 The quadratic equation for the i^{th} spline: $f_i(x) = ax^2 + bx + c$

4 polynomials with 3 coefficients $\Rightarrow 12$ coefficients to be determined
 $a_1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3, a_4, b_4, c_4$
 $a_1 = 0 \Rightarrow$ other 11 determined from 11 equations

- 1 $i=1 \quad f_1(x) = a_1x^2 + b_1x + c_1 = b_1x + c_1 = 5$
- 2 $f_1(x) = a_1x^2 + b_1x + c_1 = b_1 \cdot 11 + c_1 = 9$
- 3 $i=2 \quad f_2(x) = a_2x^2 + b_2x + c_2 = a_2 \cdot 11^2 + b_2 \cdot 11 + c_2 = 9$
- 4 $f_2(x) = a_2x^2 + b_2x + c_2 = a_2 \cdot 15^2 + b_2 \cdot 15 + c_2 = 10$
- 5 $i=3 \quad f_3(x) = a_3x^2 + b_3x + c_3 = a_3 \cdot 15^2 + b_3 \cdot 15 + c_3 = 10$
- 6 $f_3(x) = a_3x^2 + b_3x + c_3 = a_3 \cdot 18^2 + b_3 \cdot 18 + c_3 = 8$
- 7 $i=4 \quad f_4(x) = a_4x^2 + b_4x + c_4 = a_4 \cdot 18^2 + b_4 \cdot 18 + c_4 = 8$
- 8 $f_4(x) = a_4x^2 + b_4x + c_4 = a_4 \cdot 22^2 + b_4 \cdot 22 + c_4 = 7$

Other 3 equations are obtained from condition that ^{at the} interior knots the slopes (first derivatives) of the polynomials for adjacent intervals are equal:

- 9 $i=2: 2ax_2 + b_1 = 2a_2x_2 + b_2 \Rightarrow b_1 = 2a_2 \cdot 11 + b_2$
- 10 $i=3: 2ax_3 + b_2 = 2a_3x_3 + b_3 \Rightarrow 2a_2 \cdot 15 + b_2 = 2a_3 \cdot 15 + b_3 = 0$
- 11 $i=4: 2ax_4 + b_3 = 2a_4x_4 + b_4 \Rightarrow 2a_3 \cdot 18 + b_3 = 2a_4 \cdot 18 + b_4 = 0$

Spline Interpolation

8	1	0	0	0	0	0	0	0	0	0	b_1	5
11	1	0	0	0	0	0	0	0	0	0	c_1	9
0	0	11^2	11	1	0	0	0	0	0	0	a_2	9
0	0	15^2	15	1	0	0	0	0	0	0	b_2	10
0	0	0	0	0	15^2	15	1	0	0	0	$c_2 =$	10
0	0	0	0	0	18^2	18	1	0	0	0	a_3	8
0	0	0	0	0	0	0	0	18^2	18	1	b_3	8
0	0	0	0	0	0	0	0	22^2	22	1	c_3	7
1	0	-22	-1	0	0	0	0	0	0	0	a_4	0
0	0	30	1	0	-30	-1	0	0	0	0	b_4	0
0	0	0	0	0	36	1	0	-36	-1	0	c_4	0